

Planning and Policy Selection in Regime-Dependent Active Inference

Mathematical Memorandum IIIB

Prepared for Dr. Martin Bello

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Abstract

This memorandum develops the planning consequences of the regime-dependent generative models introduced in Mathematical Memorandum IIIA. The central result is conservative: the expected free-energy objective, variational free energy, preference distributions, policy precision, and Bayesian inference over policies are preserved. The extension occurs in the predictive model. Instead of predicting only future continuous hidden states, the agent predicts joint trajectories over continuous latent states and discrete operating regimes. Candidate policies therefore induce probability distributions over hybrid trajectories of the form (x_k, M_k) , where x_k denotes the continuous latent affective-cognitive-interoceptive state and M_k denotes the inferred operating regime. Expected free energy is computed using these enriched predictive distributions, but its formal role is unchanged. This provides a mathematically conservative formulation of planning in regime-dependent Active Inference and prepares the application to the addictive cycle developed in Memorandum IIIC.

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1 Purpose and relation to the previous memoranda

The preceding documents established three complementary foundations. The conceptual white paper proposed that the addictive cycle is better understood as a progression through qualitatively distinct operating regimes rather than as a single stationary maladaptive process. Mathematical Memorandum II reformulated Active Inference in explicit stochastic state-space and finite-horizon control notation. Mathematical Memorandum IIIA then introduced the minimal additional architecture required to represent regime switching: a continuous latent state x_k , an emergent dysregulation functional $D(x_k; \theta)$, a discrete regime variable M_k , regime-dependent generative models $\mathcal{G}_M$, and transition probabilities over regimes.

The present memorandum answers a narrower question:

How does Active Inference planning change when the predictive generative model is regime dependent?

The answer is deliberately modest. The optimization principles of Active Inference do not change. Variational free energy remains the objective for approximate inference over hidden states. Expected free energy remains the objective for evaluating future policies. Preference distributions continue to replace ordinary reward functions. Policy precision continues to modulate confidence in policy selection. Bayesian inference over policies is preserved.

The only modification is that future predictive distributions must now be defined over both continuous latent states and discrete operating regimes. Planning therefore becomes prediction over hybrid continuous-discrete trajectories. This is the central idea of the memorandum.

2 Notation

Table 1 summarizes the principal notation used throughout this document.

3 Standard trajectory-based Active Inference planning

We briefly recall the trajectory notation from Mathematical Memorandum II. In the standard continuous-state formulation, the latent dynamics and observation model may be written as

$$x_{k+1} = f(x_k, u_k, w_k), \quad (1)$$

$$y_k = h(x_k, v_k), \quad (2)$$

where w_k denotes process noise and v_k denotes observation noise. Equivalently, the model is specified by transition and observation densities

$$p(x_{k+1} \mid x_k, u_k), \quad (3)$$

$$p(y_k \mid x_k). \quad (4)$$

At time t , let the planning horizon be H . A policy π induces a future control sequence

$$U_\pi = [u_t, u_{t+1}, \dots, u_{t+H-1}]^T. \quad (5)$$

Symbol	Meaning
$x_k \in \mathbb{R}^n$	Continuous hidden state at time k . In the addiction application, this may include fear, anger, reality testing, craving, euphoric recall, shame, isolation, policy precision, and interoceptive precision.
$y_k \in \mathbb{R}^m$	Observation at time k .
u_k	Action or intervention at time k .
$M_k \in \mathcal{M}$	Discrete operating regime at time k , with $\mathcal{M} = \{M_1, \dots, M_5\}$ in the addictive-cycle application.
$D(x_k; \theta)$	Emotional dysregulation functional computed from the continuous latent state.
$\mathcal{G}_M$	Regime-specific generative model indexed by regime M .
π	Candidate policy.
U_π	Future control sequence induced by policy π .
$q_\pi(x_k, M_k)$	Predictive distribution over future continuous states and regimes under policy π .
$G(\pi)$	Expected free energy of policy π .
$F(\pi)$	Variational free-energy contribution associated with policy π after observations are obtained.
γ	Policy precision or inverse temperature.

Table 1: Principal notation for planning in regime-dependent Active Inference.

The corresponding future continuous state trajectory is

$$X_\pi = [x_{t+1}, x_{t+2}, \dots, x_{t+H}]^T, \quad (6)$$

and the predicted observation trajectory is

$$Y_\pi = [y_{t+1}, y_{t+2}, \dots, y_{t+H}]^T. \quad (7)$$

The predictive state density under policy π evolves recursively as

$$q_\pi(x_{k+1}) = \int p(x_{k+1} \mid x_k, u_k) q_\pi(x_k) dx_k, \quad (8)$$

with predicted observations

$$q_\pi(y_k) = \int p(y_k \mid x_k) q_\pi(x_k) dx_k. \quad (9)$$

Expected free energy evaluates candidate policies using these predictive distributions. A useful preference-plus-information form is

$$G(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_\pi(y_k)} [-\log p_{\text{pref}}(y_k)] - \sum_{k=t+1}^{t+H} I_k(\pi), \quad (10)$$

where

$$I_k(\pi) = \mathbb{E}_{q_\pi(y_k)} [D_{\text{KL}}(q_\pi(x_k \mid y_k) \parallel q_\pi(x_k))] \quad (11)$$

is the expected information gain from future observation y_k .

The risk-ambiguity form may also be written as

$$G(\pi) = \sum_{k=t+1}^{t+H} \left\{ D_{\text{KL}}(q_\pi(y_k) \parallel p_{\text{pref}}(y_k)) + \mathbb{E}_{q_\pi(x_k)} [\mathcal{H}(p(y_k \mid x_k))] \right\}. \quad (12)$$

Policy selection is then expressed as Bayesian inference over policies:

$$q(\pi_i) = \frac{\exp[-F(\pi_i) - \gamma G(\pi_i)]}{\sum_j \exp[-F(\pi_j) - \gamma G(\pi_j)]}. \quad (13)$$

The present memorandum does not alter Equations (10)-(13). Instead, it replaces the predictive distribution over future states with a predictive distribution over future states and regimes.

4 Regime-dependent predictive dynamics

Mathematical Memorandum IIIA introduced a discrete operating regime

$$M_k \in \mathcal{M}, \quad \mathcal{M} = \{M_1, \dots, M_R\}, \quad (14)$$

where $R = 5$ in the addictive-cycle application. Each regime indexes a regime-specific generative model

$$\mathcal{G}_M = (f_M, h_M, p_M, \gamma_M, \lambda_M, \Pi_M), \quad (15)$$

where the components denote, respectively, the regime-specific latent dynamics, observation mapping, priors or preferences, policy precision, interoceptive or social-affective precision, and admissible policy landscape.

The regime-dependent latent dynamics are

$$x_{k+1} = f_{M_k}(x_k, u_k, w_k), \quad (16)$$

or, equivalently,

$$p(x_{k+1} \mid x_k, u_k, M_k). \quad (17)$$

The observation model may also depend on the active regime:

$$y_k = h_{M_k}(x_k, v_k), \quad (18)$$

or

$$p(y_k \mid x_k, M_k). \quad (19)$$

For the most conservative first formulation, it is sufficient to let only the transition dynamics depend on M_k . However, Equation (15) permits a more general interpretation in which the observation mapping, prior preferences, precision parameters, and admissible policies may also vary by regime. This does not introduce a new optimization principle. It only specifies what may vary inside the generative model.

Regime evolution is governed by a transition law

$$P(M_{k+1} \mid M_k, x_k, u_k). \quad (20)$$

The transition law may depend on the dysregulation functional

$$D_k = D(x_k; \theta), \quad (21)$$

so that high dysregulation increases the probability of transitions into less adaptive regimes, while successful regulatory actions increase the probability of returning to more adaptive regimes.

5 Joint predictive distributions over states and regimes

In standard Active Inference planning, a policy induces $q_\pi(x_k)$. In the regime-dependent formulation, a policy induces

$$q_\pi(x_k, M_k). \quad (22)$$

This joint predictive distribution is the central mathematical object of regime-dependent planning. It encodes uncertainty about both the future continuous latent state and the future operating regime.

Given a current belief $q_t(x_t, M_t)$, the one-step predictive distribution under action u_k is

$$q_\pi(x_{k+1}, M_{k+1}) = \sum_{M_k \in \mathcal{M}} \int p(x_{k+1} \mid x_k, u_k, M_k) P(M_{k+1} \mid M_k, x_k, u_k) q_\pi(x_k, M_k) dx_k. \quad (23)$$

The predicted observation distribution becomes

$$q_\pi(y_k) = \sum_{M_k \in \mathcal{M}} \int p(y_k \mid x_k, M_k) q_\pi(x_k, M_k) dx_k. \quad (24)$$

Equations (23) and (24) replace Equations (8) and (9). The expected free-energy objective itself is unchanged; only the distributions used to compute it are enriched.

The marginal predictive distribution over continuous states is recovered by summing over regimes:

$$q_\pi(x_k) = \sum_{M_k \in \mathcal{M}} q_\pi(x_k, M_k). \quad (25)$$

Similarly, the predictive distribution over regimes is

$$q_\pi(M_k) = \int q_\pi(x_k, M_k) dx_k. \quad (26)$$

These marginals are useful diagnostically, but the joint distribution $q_\pi(x_k, M_k)$ is the object needed for planning because the future evolution of x_k depends on M_k .

6 Hybrid trajectories

The standard future trajectory X_π is now replaced by a hybrid continuous-discrete trajectory

$$\mathcal{T}_\pi = \{(x_{t+1}, M_{t+1}), \dots, (x_{t+H}, M_{t+H})\}. \quad (27)$$

The associated future observations remain

$$Y_\pi = [y_{t+1}, \dots, y_{t+H}]^T, \quad (28)$$

but their distribution is now generated by the joint predictive law over (x_k, M_k) .

A policy therefore induces a distribution over hybrid trajectories:

$$q_\pi(\mathcal{T}_\pi) = q_\pi((x_{t+1}, M_{t+1}), \dots, (x_{t+H}, M_{t+H})). \quad (29)$$

This distribution includes uncertainty about both continuous state evolution and possible future regime switches. In the addition application, a candidate policy may therefore be evaluated not only by its predicted effect on craving or shame, but also by its predicted effect on the probability of remaining in recovery, entering emotional destabilization, progressing into cognitive capture, or collapsing into an autopilot-like acting-out regime.

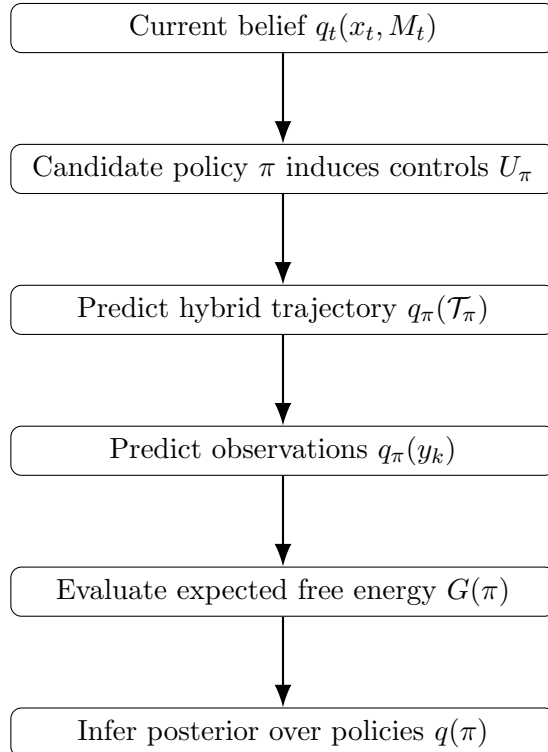


Figure 1: Planning in regime-dependent Active Inference. The Active Inference objective is unchanged, but policy evaluation proceeds through predictive distributions over hybrid continuous-discrete trajectories.

7 Expected free energy under regime-dependent prediction

The preference-plus-information form of expected free energy remains

$$G(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_\pi(y_k)} [-\log p_{\text{pref}}(y_k)] - \sum_{k=t+1}^{t+H} I_k(\pi). \quad (30)$$

The difference is that $q_\pi(y_k)$ is now computed from Equation (24). Thus the predicted observation distribution already incorporates uncertainty about future regimes.

The expected information gain term is generalized from information about x_k alone to information about the joint hidden state (x_k, M_k) :

$$I_k^{\text{hyb}}(\pi) = \mathbb{E}_{q_\pi(y_k)} [D_{\text{KL}}(q_\pi(x_k, M_k \mid y_k) \parallel q_\pi(x_k, M_k))]. \quad (31)$$

Using this term, the hybrid preference-plus-information form is

$$G(\pi) = \sum_{k=t+1}^{t+H} \mathbb{E}_{q_\pi(y_k)} [-\log p_{\text{pref}}(y_k)] - \sum_{k=t+1}^{t+H} I_k^{\text{hyb}}(\pi). \quad (32)$$

Equation (32) does not define a new free energy. It is the same expected free-energy functional evaluated under a richer hidden-state space. The hidden state now contains both continuous and discrete components.

Equivalently, a risk-ambiguity form may be written as

$$G(\pi) = \sum_{k=t+1}^{t+H} \left\{ D_{\text{KL}}(q_\pi(y_k) \parallel p_{\text{pref}}(y_k)) + \sum_{M_k \in \mathcal{M}} \int q_\pi(x_k, M_k) \mathcal{H}(p(y_k \mid x_k, M_k)) dx_k \right\}. \quad (33)$$

The first term penalizes predicted observations that diverge from preferred observations. The second term penalizes predicted states and regimes that generate ambiguous observations. Again, the formal interpretation is unchanged; only the predictive distribution has been generalized.

8 Regime-dependent preferences and policy landscapes

The most conservative implementation keeps the preference distribution fixed:

$$p_{\text{pref}}(y_k \mid M_k) = p_{\text{pref}}(y_k). \quad (34)$$

Under this assumption, regimes affect planning through state transitions, observation predictions, and precision structure, but not through explicit changes in preferred outcomes.

A more general formulation permits regime-dependent preferences:

$$p_{\text{pref}}(y_k \mid M_k). \quad (35)$$

This does not imply that the biological system has adopted a different normative objective in any philosophical sense. Rather, it permits the generative model to encode different expected or sought observations in different operating modes. For example, in recovery a preferred future observation might correspond to safety, connection, and reduced craving. In cognitive capture, the same system may assign excessive expected salience to observations associated with acting out.

When preferences depend explicitly on regimes, the extrinsic component of expected free energy becomes

$$G_{\text{pref}}(\pi) = \sum_{k=t+1}^{t+H} \sum_{M_k \in \mathcal{M}} \int q_{\pi}(y_k, M_k) [-\log p_{\text{pref}}(y_k \mid M_k)] dy_k. \quad (36)$$

Similarly, the admissible policy set may depend on regime:

$$\pi \in \Pi_{M_k}. \quad (37)$$

This is useful for modeling policy narrowing. In adaptive regimes, the agent may have access to a broad set of recovery-oriented policies. In destabilized or trance-like regimes, the available policy landscape may become severely restricted. The active policies are not merely selected differently; the set of policies represented as feasible may itself change.

For planning over a horizon, it may be preferable to define a policy as an object conditioned on possible future regimes:

$$\pi = \{\mu_M : M \in \mathcal{M}\}, \quad (38)$$

where μ_M maps beliefs to actions when regime M is active. This interpretation is close to a feedback policy in stochastic control. A simpler first-pass implementation may instead evaluate open-loop control sequences U_{π} , as in Memorandum II.

9 Policy precision and posterior policy selection

The posterior over policies retains the standard softmax form:

$$q(\pi_i) = \frac{\exp[-F(\pi_i) - \gamma G(\pi_i)]}{\sum_j \exp[-F(\pi_j) - \gamma G(\pi_j)]}. \quad (39)$$

The policy precision γ may be treated in several ways. The simplest approach uses a fixed precision parameter. A state-dependent approach uses

$$\gamma_k = \gamma(x_k), \quad (40)$$

while a regime-dependent approach uses

$$\gamma_k = \gamma_{M_k}. \quad (41)$$

The design specification for Memorandum IIIA included γ_k as a coordinate of the continuous latent state. In that case, policy precision may be inferred dynamically as part of x_k . This is compatible with the regime-dependent formulation because the evolution of γ_k may depend on M_k :

$$\gamma_{k+1} = f_{M_k}^{(\gamma)}(x_k, u_k, w_k), \quad (42)$$

where $f_{M_k}^{(\gamma)}$ denotes the component of the regime-specific dynamics governing policy precision.

In addiction modeling, altered precision is clinically important. During recovery, policy precision may support stable commitment to adaptive actions. During cognitive capture, precision may become concentrated around addictive policies. During trance or autopilot-like states, effective precision may collapse around a narrow compulsive action sequence. These are not different optimization principles. They are different precision structures inside the generative model.

10 Planning as prediction of relapse and recovery pathways

The regime-dependent planning framework permits a policy to be evaluated by its effect on predicted regime trajectories. For example, in the five-regime addictive-cycle model,

$$\mathcal{M} = \{M_1, M_2, M_3, M_4, M_5\}, \quad (43)$$

where the regimes correspond to Wise Mind/Recovery, Emotional Destabilization, Addictive Cognitive Capture, Trance/Policy Collapse, and Post-Acting-Out Aftermath.

A candidate recovery policy may be evaluated in part by the predicted probability of remaining in, or returning to, M_1 :

$$P_\pi(M_{t+H} = M_1) = \int q_\pi(x_{t+H}, M_{t+H} = M_1) dx_{t+H}. \quad (44)$$

Likewise, a relapse-risk measure may be defined as the predicted probability of entering M_4 within the planning horizon:

$$R_{\text{relapse}}(\pi) = P_\pi(\exists k \in \{t+1, \dots, t+H\} : M_k = M_4). \quad (45)$$

Equation (45) is not an additional Active Inference objective. It is a diagnostic summary derived from the same predictive distribution $q_\pi(\mathcal{T}_\pi)$. Such quantities may be useful for interpretation, simulation, or clinical decision support, but they should not be confused with expected free energy itself.

The dominant relapse pathway can be represented as

$$M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow M_4 \rightarrow M_5, \quad (46)$$

while recovery-oriented transitions may include

$$M_2 \rightarrow M_1, \quad M_3 \rightarrow M_2, \quad M_5 \rightarrow M_1. \quad (47)$$

Aftermath feedback is clinically important:

$$M_5 \rightarrow M_2, \quad M_5 \rightarrow M_3, \quad (48)$$

because shame, guilt, despair, and isolation can reinitiate emotional destabilization or addictive cognitive capture.

Within the planning formulation, such pathways are not imposed as deterministic sequences. They appear as high-probability paths under the transition law $P(M_{k+1} | M_k, x_k, u_k)$, conditioned by the current latent state, dysregulation level, and actions taken.

11 The conservative character of the extension

It is useful to state explicitly what has and has not changed.

Component	Status in regime-dependent planning
Variational free energy F	Preserved. Used for approximate inference over hidden causes.
Expected free energy $G(\pi)$	Preserved. Evaluated under joint predictive distributions over states and regimes.
Preference distributions	Preserved. May be fixed or regime dependent, depending on modeling choice.
Policy precision	Preserved. May be fixed, state dependent, regime dependent, or treated as a latent coordinate.
Bayesian policy inference	Preserved. Softmax posterior over policies remains unchanged.
Predictive model	Enriched. Predictions are over hybrid trajectories (x_k, M_k) .
Optimization principle	Unchanged. The proposal modifies the generative model, not the principle of free-energy minimization.

Table 2: What changes and what remains fixed in regime-dependent Active Inference planning.

The central scientific point is therefore:

Regime-dependent Active Inference does not replace the expected free-energy objective.
It changes the predictive generative model over which expected free energy is evaluated.

This distinction is essential. Addiction is not modeled as a violation of Active Inference. It is modeled as inference and planning under a generative model whose operating regime, latent dynamics, precision structure, and policy landscape have shifted.

12 Open modeling choices

Several modeling choices remain open and should be treated as research questions rather than settled assumptions.

1. Should the observation model $p(y_k | x_k, M_k)$ be regime dependent, or should regime dependence initially be restricted to the latent transition law?
2. Should preference distributions be fixed across regimes, or should they be allowed to vary with M_k ?
3. Should policy precision γ_k be modeled as a latent state variable, a regime-dependent parameter, or both?

4. Should policies be represented as open-loop control sequences U_π , feedback policies $\mu(b_k)$, or regime-conditioned contingencies $\{\mu_M\}_{M \in \mathcal{M}}$?
5. Can the regime transition law $P(M_{k+1} \mid M_k, x_k, u_k)$ be learned from longitudinal behavioral, physiological, or self-report data?
6. Should quantities such as relapse risk $R_{\text{relapse}}(\pi)$ remain diagnostic summaries, or should they eventually enter preference distributions explicitly?
7. How should the hybrid information-gain term $I_k^{\text{hyb}}(\pi)$ be approximated computationally in realistic models?

These questions are not weaknesses of the formulation. They identify the degrees of freedom that must be resolved when moving from a mathematical architecture to a concrete computational model.

13 Conclusion

This memorandum developed the planning consequences of regime-dependent generative models. The core modification is simple: standard Active Inference predicts future continuous hidden states, whereas regime-dependent Active Inference predicts future continuous hidden states together with discrete operating regimes.

The central predictive object is therefore

$$q_\pi(x_k, M_k), \quad (49)$$

and the corresponding future trajectory is

$$\mathcal{T}_\pi = \{(x_{t+1}, M_{t+1}), \dots, (x_{t+H}, M_{t+H})\}. \quad (50)$$

Expected free energy, variational free energy, preference distributions, policy precision, and Bayesian policy inference remain intact. The enriched predictive model allows policies to be evaluated not only by their expected future observations, but also by their predicted effects on regime transitions.

In the addiction application, this means that relapse and recovery pathways can be treated as inferred future trajectories of the generative model rather than as external labels imposed after the fact. The agent plans under uncertainty about both its future affective-cognitive state and its future operating regime. This provides the mathematical bridge from the regime-dependent generative models of Memorandum IIIA to the addictive-cycle application developed in Memorandum IIIC.

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